Research GPT 5.6 Terra

Second-round analysis: mechanistic estimation of the quenched deep third cumulant

Scope. This report uses only the second-round measurements supplied in the brief. Closed mechanisms are not re-proposed. The key new constraint is the budget window: V29 is about 250 units, its old-source tier is 107, so the non-old cost is about 143 units; leaders sit at 133-165, leaving at most about 22 units for all deep content.

Bottom line

[DERIVED] The "aggregate response state" survives the logical tests, but its stated implementation is only budget-compatible at roughly k = 1, not k = 4, 8, 32, etc. With about 11-12 old-state updates, 2k units/update gives about 22-24k total units. A second component already costs roughly 44-48 units and misses the leader window.

[CONJECTURE] The most credible route is therefore not generic tensor compression. It is an unusually strong one- or two-component aggregate response law whose components are fitted in the observable D21/D3 query metric at the age gate and then transported without source identity.

[MEASURE-THIS] The first decisive experiment is simply whether k=1 reaches 2% query error on the stored age-gate legs. Everything else is secondary until this is known.

Q1. Concrete fitting procedure for the aggregate response state

Mathematical object. At the age gate, replace the sum of deep source tensors by

```
K_old ≈ Σ(a=1..k) Sym3(G_a ⊗ v_a), G_a = G_a^T, G_a ∈ R^(n×n), v_a ∈ R^n.
```

```
Its observable slices are
```

```
D21_ic = Σ_a [ (G_a)_ii (v_a)_c + 2 (G_a)_ic (v_a)_i ], D3_i = 3 Σ_a (G_a)_ii (v_a)_i.
```

The fit should not minimize tensor Frobenius error. It should minimize the actual query family induced by the four stored slice contractions. Write Φ_t(K) for the t-th sibling/query (U and its three siblings). Solve

```
min_{G_a,v_a} Σ_t || Φ_t(K_exact) - Φ_t(K_k) ||_F^2 / ||Φ_t(K_exact)||_F^2, target ≤
0.02^2.
```

Practical fitting algorithm

• [MEASURE] Use one stored age-gate .npz (prefer layer 6 or 10) with exact dense A_s, P_s and w2. Build the exact four query outputs only once; do not form a third-order tensor.

• [CONJECTURE] Evaluate the objective with 64-128 Gaussian Hutchinson probes per output: for an error matrix E, E||Eg||² = ||E||_F². This avoids repeatedly storing full residuals while preserving a Frobenius target.

• [CONJECTURE] Alternate variables. For fixed v_a, the map G_a -> D21 is linear, so solve a symmetric least-squares problem. For fixed G_a, update v_a by linear least squares. Re-orthogonalize between components and accept a component only when it reduces the stacked query residual.

• [MEASURE] Run k = 1, 2, 4, 8 with 2-3 random restarts. Stop when the held-out probe error crosses 2%. Use 25% of probes as validation so the fit cannot overfit its sketch.

Cost

Transport is exact under B = W diag(w1): G_a <- B G_a B^T and v_a <- B v_a. Under the supplied metering this is about 2 units/component/layer for the dense G update; vector transport is negligible by comparison. Therefore

```
Cost_old ≈ 2 k N_old units, with N_old ≈ 11-12.
```

This gives about 22-24 units for k=1, 44-48 for k=2, and 88-96 for k=4. Since the leader window leaves at most about 22 units above the 143-unit non-old bill, k=1 is the only directly budget-compatible case; k=2

needs an additional factor-of-two transport optimization.

What k should we expect?

[DERIVED] M3 says the exact observable U is rank about 100 for 99% energy and about 230 for a 2% target. That does not imply k ≈ 100 or 230 here: one Sym3(G⊗v) can generate a full-rank D21 matrix because the term 2 diag(v)G can have rank n.

[CONJECTURE] Expressivity therefore does not rule out k=1 or k=2. But the measured flat spectra make k=1 an aggressive hypothesis rather than a default expectation. My practical prior is k=1-2; the budget, however, makes k=1 the only result that could immediately explain the leaders.

First falsifying experiment

[MEASURE-THIS] Fit the four query siblings at one gate (layer 6 or 10). Report held-out relative RMS error in U, D3, and the other three siblings. If k=1 > 2% on the held-out probes, reject the stated aggregate-response mechanism as an explanation of the 133-165-unit leaders. If k=1 passes, repeat at layer 14 and after one transport step; if it remains <2%, this becomes the leading experimental path.

Q2. Other genuinely new formulations

I would distinguish one technically new idea from several mechanisms that now fail a budget or information-content test.

A. Quenched KL / Wiener-chaos state - technically new, but likely too expensive

Object. Represent the realized old tensor as K_old ≈ Σ_{a=1}^q ξ_a Ψ_a(C), where Ψ_a are conditional covariance/Karhunen-Loève modes of the odd quenched field and ξ_a are the realized odd coefficients. This is not expectation-only: it explicitly keeps the random sign realization.

Update. Propagate the mode tensors under the linear B action and rotate/renormalize coefficients ξ. The even covariance controls the mode metric; the actual ξ carry the odd quenched content.

Cost. If each mode requires a dense matrix-side transport, the cost is again about 2q units/layer. Even a modest q=8 is ~176-192 units over 11-12 layers, already exceeding the entire leader headroom. A vector-only implementation would be cheaper, but then it is effectively the aggregate-response state in another coordinate system.

First falsifier. Randomized Lanczos on the conditional covariance of the observable D21 field: if the first q≤2 modes do not capture 98% of the query energy at the gate, this cannot explain the leaders.

B. Characteristic-function / random-feature basis - reject

Object/update. A low-dimensional expansion of the quenched cumulant-generating functional, with coefficients transported by B and evaluated through a small feature bank. Cost: q feature channels need at least O(q n^2) work per layer; if each channel needs a matrix transport it is closer to 2q units/layer. M3 already shows random TensorSketch features require variance corresponding to roughly 10^8 features for 2% in the tested quadratic query, so the natural random-feature route is numerically closed.

First falsifier. On the stored gate, sweep q=16,32,64,128 random Fourier/Hermite features and measure the same held-out 2% query metric. Any q that passes but costs >22 total old units cannot explain the leaders anyway.

C. Response-function / Jacobian-statistics state - reject as a leader explanation

Object. Carry a small family of response functions R_a that predicts all future D21/D3 queries without representing K_old explicitly. Update: R_a <- R_a ∘ B and re-project to a q-dimensional response basis. Cost: a vector response family can be cheap, but a matrix-valued response family costs about q units/layer; to reproduce a generic observable matrix with rank near 230 at 2% accuracy needs q of that order unless a new algebraic collapse is found. That is far outside the 22-unit headroom.

Why this is not the dead adjoint idea: a single final adjoint is already closed; this would require a whole low-dimensional future-observability subspace. M3 gives no evidence that such a subspace is tiny.

D. Contract the birth index j first - reject on arithmetic

For A=QF_A and P=QF_P, changing contraction order creates the core T_abc = Σ_j w_j F_A,aj F_P,bj F_A,cj. Generic formation is O(n r^3), about 108 units for r=384; alternatively, any exact method must pay at least the supplied 2 n^2 r lower bound, about 0.375 unit/source-layer. With roughly 10 old sources this optimistic floor is already ~3.75 units/layer, or about 41-45 units over 11-12 layers. Thus even an unrealistically perfect lower-bound implementation does not fit the leader window unless it also merges source identities.

First falsifier. Search for a shared low-rank factorization of the birth-index core that reduces all old sources jointly to ≤1 effective component/layer. If not, the contraction-order route cannot hit 133-165 units.

E. TAP / cavity / Feynman-diagram resummation - reject as presently unsupported

Object/update. A closed set of local order parameters or resummed diagrams carrying quenched third-order structure. Cost: O(k n^2) per layer would be cheap for small k, but no closed recursion for the required D21 channel follows from the supplied evidence. Parity says the relevant odd part has zero conditional mean, so a low-order even cavity closure cannot recover the realized sign pattern by itself.

First falsifier. Use the stored exact K3 responses to estimate the block-Hankel rank of future D21/D3 observations. If the observable state dimension exceeds ~2-4, any O(1)-parameter TAP explanation is ruled out at the required 2% level.

Q3. Most likely leader mechanism

[CONJECTURE] The leaders most likely use a global, source-agnostic response state that merges deep births at the age gate, with an observable parameterization so compressed that only about one dense matrix transport per deep layer is required. In other words, they probably do not "compress each old source"; they have removed source identity from the representation before the expensive old tier begins.

Why this fits the measurements: every birth matters (M2), every late layer matters (M2), parity kills mean-field averaging, M1 kills low-rank factorization of the visible Hadamard products, and M3 says the final query object itself has moderate rank. The remaining way to remove ~107 units without losing quenched information is to change the state variable so that many source histories become one small response state.

Observable that would confirm it. Measure leader cost versus the number of old births presented to the method. A source-identity method should grow approximately linearly with old-source count. A true aggregate-response method should show an almost flat old-tier cost once the age gate is crossed, while accuracy stays near the 2e-8 regime. A second confirmation is a local fit in which k=1 reproduces the four age-gate query families to ≤2% and remains stable after transport.

Strongest conclusion if k=1 fails

[DERIVED FROM THE CURRENT EVIDENCE] If k=1 fails the 2% gate test, and k=2 is needed, then the stated aggregate response implementation cannot explain the public leader bill because its transport alone is ~44-48 units. Combined with parity, the full-rank Hadamard spectra, the weak random-feature sketches, the source/layer sensitivity, and the arithmetic lower bounds on exact contraction, the evidence would then support the stronger conclusion that no currently specified mechanism is plausibly capable of reaching 133-165 units. At that point the single most informative next measurement would be a direct cost decomposition of the leaders via controlled local perturbations: increase only the number of deep-source identities or only the depth of the old regime and fit the resulting submission cost curve.

End of second-round report